

ADITYA KULKARNI

(281) 876-7066

✉ adityakulkarnius@gmail.com [in linkedin.com/in/aditya-kulkarni-355b81217](https://www.linkedin.com/in/aditya-kulkarni-355b81217) github.com/Aditya-k24 kulkarniaditya.com

EDUCATION

North Carolina State University

M.S. Computer Science

Raleigh, NC

Aug 2024 – May 2026

University of Mumbai

B.Tech. Computer Science, CGPA: 9.32/10

Mumbai, India

Aug 2020 – May 2024

WORK EXPERIENCE

Product Engineer Intern

May 2025 – Present

Isomer AI

Remote

- Shipped 10+ production backend features for an AI-powered financial automation platform – Temporal.io workflow orchestration, RBAC, real-time event pipelines (Pusher), and REST APIs – across 7 deployments using TypeScript, Node.js, Express, PostgreSQL, and Redis
- Integrated multi-model LLM tooling (Claude, GPT-4o, Gemini) with MCP and built document processing pipelines (PDF generation, Microsoft Graph API, S3); contributed 394 commits and led code review using AI-assisted development (Claude Code, Copilot)

AI/ML Intern

Jun 2023 – Aug 2023

Capgemini

Remote

- Designed and implemented distributed data pipelines using Hadoop and Spark for large-scale batch processing, and integrated Kafka streaming for real-time data ingestion – reducing ticket-processing latency by 20%
- Led end-to-end SDLC for a data-driven support ticket analysis system using NLP and ML, reducing manual processing time by 15%

Software Developer Intern

Jun 2024 – Aug 2024

Mphasis (HP Company)

Remote

- Built 5+ AI-driven automation workflows including an OCR and NLP-powered claim processing pipeline improving accuracy 25%; automated CI/CD deployments on AWS and GCP with Grafana monitoring

PERSONAL PROJECTS

VectorLift – Semantic Search & Ranking Engine | Python, FastAPI, Kafka, Elasticsearch, Qdrant, PostgreSQL 2026

- Built a production-grade semantic search system on MS MARCO: Kafka-driven indexing pipeline for event-based corpus ingestion into Elasticsearch and Qdrant; FastAPI search API with per-stage latency tracking and Streamlit experiment dashboard
- Evaluated 6 pipeline configurations (BM25, bi-encoder retrieval, cross-encoder re-ranker) with NDCG@10, MRR@10, and MAP; applied paired bootstrap significance testing across fine-tuned sentence-transformers and MiniLM re-rankers

PulseRank – Real-Time Leaderboard & Activity Feed | TypeScript, NestJS, Next.js, MySQL, MongoDB, Redis 2026

- Built an event-driven leaderboard and activity feed: MySQL for transactional writes with idempotency-key deduplication and outbox pattern, MongoDB for feed projections with TTL expiry, Redis sorted sets for sub-millisecond reads via Server-Sent Events
- Validated multi-store consistency with Testcontainers integration tests; benchmarked hot paths with k6; shipped CI/CD with GitHub Actions and instrumented OpenTelemetry + Prometheus for full observability

AlgoPulse – Agentic AI Platform | TypeScript, Node.js, React, Supabase, Temporal.io, Kafka 2025

- Architected an event-driven AI orchestration platform; identified API-polling bottleneck under load and pivoted to Supabase Edge Functions streaming AI reasoning loops directly to client, sustaining 2,000 async inference requests/second without state loss
- Scaled worker architecture with Temporal.io and Kafka for fault-tolerant state synchronization; built React frontend with real-time streaming updates and full-stack TypeScript codebase

TECHNICAL SKILLS

Languages: Python, TypeScript, Java, JavaScript, SQL, R, C

Full Stack: React, Node.js, NestJS, FastAPI, REST APIs, PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch

Data & Streaming: Kafka, Spark, Hadoop, Pandas, NumPy, S3, AWS, GCP

AI & DevOps: Claude Code, GitHub Copilot, MCP, OpenAI API, Docker, Kubernetes, Grafana, CI/CD